

Mastering the Command Line

Navigating Windows with CMD & PowerShell

The Core Concept

Demystifying the File System Tree

| The Blueprint: The Directory Tree

Students get confused because they can't **see** where they are. Remind them that the computer views files as an inverted tree hierarchy.

- **Root:** The starting drive letter (e.g., `C:\`).
- **Parent Directory:** The folder containing your current folder.
- **Child Directory:** Subfolders nested inside your current folder.

```
C:\ (Root)
├── Users
│   ├── Student
│   │   ├── Documents
│   │   └── Desktop
│   │       └── Project
```


| Where Am I? Checking Location

Looking Around

Before moving anywhere, students must learn how to see where they currently stand and what files are surrounding them.

PowerShell Advantage: It accepts both Windows native and Linux-style shortcut syntax shortcuts natively!

Action	Command Prompt	PowerShell
Show Current Location	Automatic in prompt	<code>Get-Location</code> (or <code>pwd</code>)
List Files & Folders	<code>dir</code>	<code>Get-ChildItem</code> (or <code>ls</code> / <code>dir</code>)

The Movement (CD)

Changing Directories Without Friction

Moving Around: The "CD" Rules



Move Forward

Go deeper into a child directory:

```
cd Documents
```



Move Backward

Go up to the parent directory:

```
cd ..
```



Go to Root

Jump all the way back to the drive root:

```
cd \
```



Pro Tip for Students: Tab Completion

Stop typing full folder names! Type the first 2-3 letters of a directory and press the **TAB** key to autocomplete the path instantly. This eliminates typos.

| Creating Folders & Files

Target Action	Command Prompt (CMD)	PowerShell
Create a New Folder	<code>mkdir MyFolder</code>	<code>New-Item -ItemType Directory MyFolder</code> (Shortcut: <code>mkdir MyFolder</code>)
Create an Empty File	<code>type nul > index.html</code>	<code>New-Item index.html</code> (Shortcut: <code>echo "" > index.html</code>)

| Copying & Moving Items

Managing Content

Once folders and files are created, moving or duplicating them around follows simple source-to-destination logic.

Syntax Rule:

```
Command [Source] [Destination]
```

Action	CMD Syntax	PowerShell Syntax
Copy a File	<pre>copy file.txt Backup\</pre>	<pre>Copy-Item file.txt Backup\ (Shortcut: <i>cp</i>)</pre>
Move/Rename	<pre>move file.txt Target\</pre>	<pre>Move-Item file.txt Target\ (Shortcut: <i>mv</i>)</pre>

Deleting Files & Folders

Action	Command Prompt (CMD)	PowerShell
Delete a File	<code>del file.txt</code>	<code>Remove-Item file.txt</code> (Shortcut: <code>rm</code>)
Delete an Empty Folder	<code>rmdir MyFolder</code>	<code>Remove-Item MyFolder</code>
Delete Folder + Contents	<code>rmdir /s /q MyFolder</code>	<code>Remove-Item -Recurse -Force MyFolder</code>

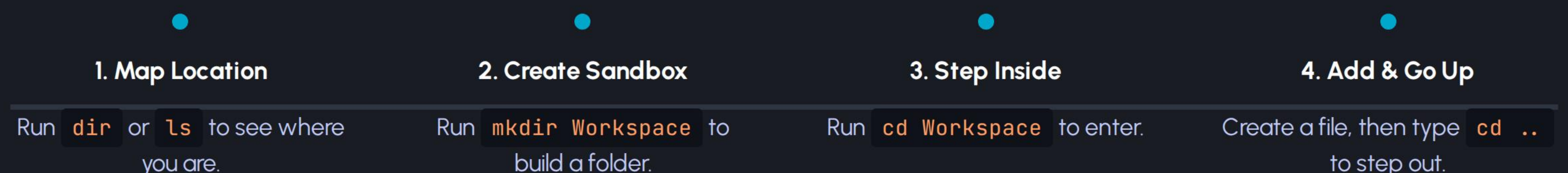
CRITICAL WARNING FOR BEGINNERS

Files deleted using Command Line interfaces **do not go to the Recycle Bin!** They are instantly, permanently erased. Double-check your spelling before pressing Enter.

The Live Cheat Sheet Matrix

Operation	CMD Standard	PowerShell Modern	Universal Linux/Mac Alias
List items	<code>dir</code>	<code>Get-ChildItem</code>	<code>ls</code> *(PowerShell only)*
Make folder	<code>mkdir</code>	<code>New-Item -Type Dir</code>	<code>mkdir</code>
Copy file	<code>copy</code>	<code>Copy-Item</code>	<code>cp</code> *(PowerShell only)*
Clear Screen	<code>cls</code>	<code>Clear-Host</code>	<code>clear</code> *(PowerShell only)*

| The Student Practice Blueprint



1. Map Location

Run `dir` or `ls` to see where you are.

2. Create Sandbox

Run `mkdir Workspace` to build a folder.

3. Step Inside

Run `cd Workspace` to enter.

4. Add & Go Up

Create a file, then type `cd ..` to step out.

Hands-On Lab Time!

Open your terminals and type along.